CCC-C-419
JULY 30, 1956
SUPERSEDING
Fed. Spec. CCC-D-771b
July 7, 1943

FEDERAL SPECIFICATION

CLOTH, COTTON, DUCK, UNBLEACHED, PLIED-YARNS (ARMY AND NUMBERED)

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope.—This specification covers the requirements of grey (unbleached) cotton cloth duck primarily intended for use in fabrication of tentage and equipage.

1.2 Classification.—The cloth, cotton, duck, plied-yarns (Army and Numbered) shall be of the following types as specified (see 6.1):

Type I.—Numbered duck
Type II.—See 6.4 and 6.5
Type III.—Army duck
Type IV.—See 6.4 and 6.5

2. APPLICABLE SPECIFICATIONS AND STANDARDS

2.1 The following specifications and standards, of the issues in effect on date of invitation for bids, form a part of this specification:

Federal Specification:

CCC-T-191—Textile Test Methods.

Federal Standard:

Standard No. 4—Glossary of Fabric Imperfections.

(Activities outside the Federal Government may obtain copies of Federal Specifications and Standards as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington 25, D. C.

(Single copies of this specification and other products specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, Seattle, and Washington, D. C.

(Federal Government activities may obtain copies of Federal Specifications and Standards and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Specification:

MIL-P-12130—Packaging of Fabrics for Tentage and Equipage.

Military Standards:

MIL-STD-105 — Sampling Procedures and Tables for Inspection by Attributes and Appendix thereto.

MIL-STD-129—Marking for Shipment and Storage.

(Copies of Military specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Material.

FSC Class 8305

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3.1.1 *Cotton.*—The cotton shall be of suitable length of staple and grade to meet the requirements of this specification.

3.1.2 *Yarn.*—The cotton used in the manufacture of the yarn shall be well carded, drawn, evenly spun, and twisted into plied yarns. The ply shall be as specified in 3.2.

3.2 Physical requirements. — Cotton cloth duck covered by this specification shall conform to the requirements specified in tables I and II when tested as specified in 4.3.

TABLE I.—*Physical requirements—type I, numbered duck*

Number of duck, commercial designation	Weight, per square yard, minimum	Yarn ply, minimum		Yarns per inch, minimum		Breaking strength, minimum		Air permeability, maximum
		Warp	Filling	Warp	Filling	Warp	Filling	
Hard texture	*Ounces*					*Pounds*	*Pounds*	*ft.³/min./ft.²*
12/0	48.00	9	9	20	12	800	430	
2/0	31.90	5	7	24	16	465	435	
1/0	30.31	5	6	24	16	450	405	
1	28.71	5	5	26	19	440	370	2
2	27.12	5	4	26	19	420	345	
3	25.53	4	5	29	20	390	330	
4	23.93	4	4	31	22	375	300	2
5	22.33	3	4	35	23	345	285	
6	20.74	3	3	35	25	335	250	2
8	17.55	3	3	43	25	285	210	2
10	14.35	3	3	45	27	245	160	4
11	12.75	3	2	45	34	240	140	
12	11.16	2	2	45	35	195	120	4
Medium texture								
1	28.71	5	5	26	15	425	345	
2	27.12	5	4	26	15	410	320	
3	25.53	4	5	29	15	370	315	
4	23.93	4	4	29	17	350	290	
5	22.33	3	4	34	18	315	285	
6	20.74	3	3	34	18	305	250	

TABLE II.—*Physical requirements—type III, Army duck*

Weight per square yard, minimum	Yarn ply, minimum		Texture Yarns per inch, minimum		Breaking strength, minimum		Air permeability, maximum
	Warp	Filling	Warp	Filling	Warp	Filling	
Ounces					*Pounds*	*Pounds*	*ft.³/min./ft.²*
8.25	2	2	60	58	125	120	
9.85	2	2	52	40	160	110	
12.29	3	2	44	34	210	130	4
14.77	3	3	44	28	235	175	
15.90	3	3	45	34	245	200	
18.48	4	4	42	28	315	200	

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3.2.1 *Width.*—The width of the cotton cloth duck shall be as specified (see 6.1). The following tolerances will be permitted:

Width of duck (inches)	*Tolerance (inch)*
40 or less	+ ¼ – ¼
Over 40 to 51 inclusive	+ ⅜ – ¼
Over 51 to 81 inclusive	+ ⅜ – ⅜
Over 81	+ ¾ – ⅜

3.2.2 *Weave.*—The weave shall be plain, one up, one down.

3.3 Nonfibrous material content.—Cotton cloth duck shall contain not more than 2.5 percent starch and protein material and not more than 5.0 percent total size, finishing, and nonfibrous materials including starch and protein when tested as specified in 4.3. The use of finishing or loading materials to increase the weight or breaking strength is prohibited.

3.4 Workmanship.—The cotton cloth duck shall conform to the quality and grade of product established by this specification. The occurrence of defects shall not exceed the levels set by the applicable acceptable quality levels.

3.5 Length.

3.5.1 *Bolts.*—When bale packing is specified, pieces 80 yards or more in length shall be delivered. Pieces less than 80 yards will be acceptable up to 10 percent of the contract quantity, provided no piece is less than 25 yards in length, and not more than two pieces less than 80 yards are included in any one bale, except where otherwise specified in the contract (see 6.2).

3.5.2 *Rolls.*—When roll packing is specified, pieces 80 yards or more in length shall be delivered. Pieces less than 80 yards will be acceptable up to 10 percent of contract quantity, provided no piece is less than 25 yards in length, and not more than two pieces less than 80 yards are included in any one roll, except where otherwise specified in the contract (see 6.2). The pieces shall be sewn together with a butted type seam on all duck No. 10 and lighter in weight, using approximately 10/3 cord cotton thread, and have approximately 9 stitches per inch. The stitching shall be placed not closer than ½ inch from the cut or raw edge of the cotton duck. On cotton duck heavier in weight than No. 10, the pieces will be lapped and not sewn together. The number of pieces in the roll will be indicated by stencil marking on the last 6 inches of the roll on the outer surface (see 6.2).

3.6 Tickets.—Unless otherwise specified, each bolt or roll and any short length pieces contained therein shall have a ticket attached to the selvage at the inner end of the cloth with not finer than 5-ply cotton string doubled to not less than 8 inches long. The ticket shall be made of not less than 20 point cardboard and the finished surface shall not be calendered or pigmented to the degree which would incur the obliteration of printed or penciled markings. The ticket shall have a reinforced eyelet for attaching the tying cord and shall be legibly printed with water insoluble ink with the following information: (See 6.1).

Nomenclature
Type-weight-width
Stock number
Contractor's name
Contract number and date
Specification number and date
Name of contracting agency
Contractor's piece number
Gross yards _____
Inspector

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Nonmilitary.

4.1.1 *Sampling.*

4.1.1.1 *Lot.*—Unless otherwise specified, all material of the same type presented at one

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TABLE III.—*Sampling for visual and dimensional inspection*

Number of yards in lot	Number of bolts or rolls in sample	Number of yards in sample piece[1]	Acceptance number (defective)	Rejection number (defective)
300 and under	2	15	0	1
301 to 500	3	25	1	2
501 to 800	5	35	1	2
801 to 1300	7	50	2	3
1301 to 3200	10	75	3	4
3201 to 8000	15	110	4	5
8001 and over	25	150	5	6

[1] Yards in the sample piece shall be divided equally among sample rolls or bolts of duck cloth and that number of yards shall be unrolled from each sample bolt or roll.

time shall be considered a lot for purposes of inspection and tests.

4.1.1.2 *Sampling for visual and dimensional inspection.*— Unless otherwise specified, a random sample of cloth shall be taken from each lot in accordance with table III for inspection for visual and dimensional characteristics.

4.1.1.3 *Sampling for tests.*—Unless otherwise specified, random samples of duck cloth shall be taken from each inspection lot in accordance with the following table:

TABLE IV.—*Sampling for tests*

Number of rolls in lot	Number of rolls in sample[1]	Acceptance number (defective)	Rejection number (defective)
8 and under	2	0	1
9 to 15	3	0	1
16 to 40	4	0	1
41 and over	5	0	1

[1] One unit of product at least 1 yard long and the full width of the duck cloth shall be taken from each sample roll.

4.1.2 *Inspection.*—Each of the samples selected in accordance with table III, or as specified, shall be visually and dimensionally inspected to verify conformance with the requirements of this specification not including tests.

4.1.3 *Tests.*—Each of the samples selected in accordance with table IV, or as specified, shall be subjected to the tests specified in 4.3.

4.1.4 *Compliance criteria.*

4.1.4.1 *Inspection.*—Any piece in the sample for inspection failing to conform to the requirements for which inspection was made, 4.1.2, shall be considered defective. If the number of defective pieces in the sample exceeds the acceptance number for that sample specified in table III, the entire lot from which the sample was taken shall be considered as having failed to meet the requirements of this specification.

4.1.4.2 *Tests.*—Any unit of product in the sample for test failing to conform to the requirements for which tests were made, 4.1.3, shall be considered defective. If the number of defective units of products in the sample for test exceeds the acceptance number for that sample specified in table IV, the entire lot of cloth duck from which the sample was taken shall be considered as having failed to meet the requirements of this specification.

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4.2 Military.

4.2.1 *Quality assurance provisions.*—The following quality assurance provisions are applicable to inspection performed by and for the Government: Quality assurance provisions for components or materials described shall be in accordance with this specification and subsidiary specifications referenced herein to the extent applicable, except that this specification shall govern in the event of conflict.

4.2.2 *Inspection for acceptance.*—Inspection shall be in accordance with Military Standard MIL–STD–105, except where otherwise indicated hereinafter.

4.2.2.1 *Examination of the end item.*—The cloth duck shall be examined to determine conformance with the requirements of this specification. Defects found during this examination shall be classified in accordance with the list specified in 4.2.2.1.1. The Government reserves the right to require examination for any defect not included herein and to classify such defects in accordance with definition contained in Military Standard MIL–STD–105.

4.2.2.1.1 *Yard-by-yard examination.*—All defects listed below shall be counted, regardless of their proximity of one to another, except where two or more defects represent a single condition of the cloth, in which case only the more serious defect shall be counted. A continuous defect shall be counted as one defect for each warp-wise yard or fraction thereof in which it occurs. The unit of product for this examination shall be 1 linear yard.

4.2.2.1.2 *Overall examination.*—The unit of product for this examination shall be the entire piece of cloth.

Glossary No. Fed. Std. No. 4	Defect	Classification	
		Major	Minor
55, 56, 90, 91, 92	Hole, cut, tear or break (including selvage)—in which more than 3 yarns are ruptured in adjoining points.	X	
78	Smash—Any.	X	
79, 89	Abrasion mark, bruise, tender or weak spot—Any	X	
11, 33	Broken or missing yarns—		
	Three or more contiguous regardless of length.	X	
	Two contiguous, 2 inches or more missing.	X	
	Two continguous, less than 2 inches or Single 2 inches or more missing.		X
9	Fine or light filling bar, light place—clearly noticeable.	X	
2	Coarse filling bar or heavy place—clearly noticeable and exceeding 2 linear inches.		X
5, 27, 29, 77	Slough off, jerked in filling, slubs, gouts, knots—more than three times the thickness of the normal yarns.		X
9, 12	Open place or crack—clearly noticeable.	X	
13, 34	Floats—Multiple ½-inch or more in combined warp and filling directions, or harness skips.	X	
	Multiple less than ½ inch in combined warp and filling directions, or single 2 inches or more in length.		X
41	Reed mark—resulting in a clear warpwise separation markedly wider than normal.		X
40	Wrong draw—½ yard or more		X
	Width narrow or wide—beyond specified tolerances for ½ yard or more.		X
49, 50	Slack or tight selvage resulting in pronounced slackness or waviness in cloth.		X
51	Scalloped selvage—identations exceeding ½ inch.		X
48	Curled or folded selvage—½ yard or more.		X
53, 54	Stringy, beaded, corded, loopy or rough selvages—½ yard or more, resulting in raised places along the selvage edge 2 or more times the normal fabric thickness, ½ yard or more.		X
74, 75, 76	Spot, stain or streak (from oil or grease)—uncleaned, covering more than 1 inch in combined directions.	X	

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Defects

Baggy, ridgy or wavy cloth—clearly noticeable
Trash, motes, seed coat fragments, etc., clearly noticeable
Rancid or otherwise objectionable odor.

4.2.2.1.3 *Length examination.*

4.2.2.1.3.1 *Individual pieces.* — The piece shall be considered defective with regard to length requirements if its actual gross length is less than specified or more than 2 yards shorter than the gross length marked on the piece ticket. The unit of product for this examination shall be one (1) bolt or roll.

4.2.2.1.3.2 *Total yardage in sample.*—The lot shall be unacceptable if the total of the actual gross length of pieces in the sample is less than the total of the gross lengths marked on piece tickets.

4.2.2.1.4 *Acceptable quality levels for examination.*

4.2.2.1.4.1 *For yard-by-yard examination (see 4.2.2.1.1).*—

Specified width 51 inches or less—0.4 major defect and 2.5 total defects per 100 units (yards)
Specified width over 51 inches—0.65 major defects and 4.0 total defects per 100 units (yards)

4.2.2.1.4.2 *For overall examination* (see *4.2.2.1.2*).—1.0 defects per 100 units (pieces).

4.2.2.1.4.3 *For length examination of individual pieces (see 4.2.2.1.3.1).*—1.0 defects per 100 units (pieces). (See method 5010, CCC–T–191.)

4.2.2.1.5 *Inspection levels.*

(a) Yard-by-yard examination, 4.2.2.1.1, and table III, level III of Military Standard MIL–STD–105. The lot size shall be expressed in terms of one (1) yard each. An approximately equal number of yards shall be examined from each bolt or roll in the sample. The number of bolts or rolls to be drawn shall be computed as follows:

$$\frac{\text{Yards in sample}}{35}$$

(b) Overall examination (4.2.2.1.2).—The bolts or rolls examined shall be those taken for yard-by-yard examination.

(c) Examination for length of individual bolts or rolls (4.2.2.1.3.1).—The pieces or rolls examined shall be those taken for yard-by-yard examination.

4.2.3 *Testing of the end item.*—The methods of testing specified in Federal Specification CCC–T–191 wherever applicable and as listed in table V shall be followed. The physical or chemical values specified in section 3 apply to the average of the number of test specimens selected for a unit of product for test purposes as specified in the applicable test methods. The unit of product for testing shall be 1½ yards full width of the cloth. For each characteristic, the acceptable quality level shall be 4.0 per 100 units, and the inspection level shall be level L–1, table VII of Appendix, dated 27 April 1955, to Military Standard MIL–STD–105. The lot size shall be expressed in terms of one (1) yard each.

Prep for delivery req added 4192

4.3 Tests. — Tests shall be conducted in accordance with the following methods described in Federal Specification CCC–T–191.

TABLE V.—*Test methods*

Characteristic	Requirement reference	Test method number
Yarn ply	Tables I and II	Visual
Width	3.2.1	5020
Yarns per inch	Tables I and II	5050
Weight	Tables I and II	5041
Breaking strength	Tables I and II	5100
Nonfibrous materials[1]	3.3	2611
Air permeability	Tables I and II	5450

[1] The amount of nonfibrous materials shall be determined with the following formula:

$$\text{Total nonfibrous material, percent} = \frac{O-S}{O} \times 100$$

$$\text{Starch and protein material, percent} = \frac{O-S-W-C}{O} \times 100$$

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5. PREPARATION FOR DELIVERY

FED-STD-102 added

5.1 Nonmilitary.

5.1.1 *Packaging and packing.* — Unless otherwise specified, cloth duck shall be packaged and packed in accordance with good commercial practice.

5.1.2 *Marking.*—Shipping containers shall be marked with the type, weight, and color if required, of the cloth duck as defined by the contract or order number under which the shipment is made, the quantity contained therein, the name of the contractor, the number of the contract or order, the name and address of the consignee and the gross weight.

5.2 Military.

5.2.1 *Packaging, packing, and marking.*—The cloth duck shall be packaged, packed and marked for domestic and overseas shipment in accordance with Military Specification MIL–P–12130, as specified (see 6.1).

6. NOTES

6.1 Ordering data.—Procurement documents should specify the following:

(a) Title, number and date of this specification.

(b) Type and weight of duck required (see 1.2 and 3.2).

(c) Width of duck required (see 3.2.1).

(d) Type of packaging, packing, and marking, when required (see section 5).

(e) Put-up requirements (bolts or rolls) (see 3.5).

(f) Provision for deductions of defects (if any).

(g) Attachment of piece ticket (see 3.6).

(h) Nonmilitary sampling (see 4.1.1).

6.2 Short pieces.—Pieces less than 80 yards in length in excess of the amount specified in 3.5 and pieces shorter in length than 25 yards may be presented to the contracting officer for consideration for end uses where short length pieces may be considered.

6.3 This specification supersedes Military Specifications:

Navy Dept. 24C8h, 15 September 1945
MIL–D–10861(QMC), 22 January 1951

6.4 Types II and IV have been deleted and will not be required or cited for procurement.

6.5 Color for other than Military agencies.—When dyed duck is required the color, colorfastness, type of dye, and a minus tolerance of 5 percent in weight and breaking strength not in addition to tolerance in tables, should be specified in the invitation for bids.

6.6 Transportation description.—The transportation description applicable to this item is:

Cloth, cotton, waterproofed (not coated), not otherwise indexed by name.
Carload minimum weight, any quantity.
Truckload minimum weight 30,000 pounds.

Patent notice.—When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data, is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use or sell any patented invention that may in any way be related thereto.

MILITARY INTEREST:

Army—M O Q
Navy—A MC Or Sh Y
Air Force.

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